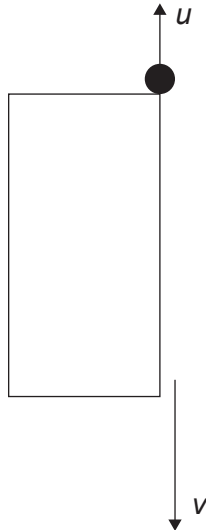


PAPER 1 / QUESTION 70

2 May 2023 / TZ1 (file label) / HL Paper 1 / original question 4 / 2223-6507

70. A stone of mass m is projected vertically upwards with speed u from the top of a cliff. The speed of the stone when it is just about to hit the ground is v .



What is the magnitude of the change in momentum of the stone?

- A. $m\left(\frac{v+u}{2}\right)$
- B. $m\left(\frac{v-u}{2}\right)$
- C. $m(v+u)$
- D. $m(v-u)$